# Chloe Barker

214-733-7514 | chloedbarkerr@icloud.com | [linkedin.com/in/chloedbarker](https://linkedin.com/in/chloedbarker) | [chloedbarker.github.io](https://chloedbarker.github.io)

## SELECT ACADEMIC PROJECTS

**MS Capstone: Groundwater Lithium & Dementia Prevalence: Texas County-Level Analysis (Python)**

- Investigated associations between naturally occurring groundwater lithium concentrations and county-level dementia prevalence across 254 Texas counties using TWDB and NORC Dementia DataHub (Medicare) data.
- Engineered exposure features (mean, median, CV, trend slope) over a 10-year window; developed an exposure homogeneity classification framework to reduce measurement error and misclassification in ecological regression.
- Built OLS regression models with HC3 heteroskedasticity-consistent standard errors; applied permutation testing (10,000 permutations) and bootstrap confidence intervals to support inference on a small homogeneous subsample.
- Conducted influence diagnostics (Cook's distance, leave-one-out analysis); 86% of LOO slopes remained in the protective direction, supporting directional robustness of findings.
- Produced choropleth maps of lithium exposure and dementia prevalence using GeoPandas and Census TIGER/Line shapefiles; identified geographic clustering in the Texas Panhandle and West Texas.

**Texas Groundwater Quality Database & Geospatial Analysis (MySQL, Python)**

- Designed and implemented a normalized MySQL relational database (64 tables, 3NF) for Texas Water Development Board groundwater-quality data across 254 counties.
- Cleaned and standardized fragmented datasets containing mineral concentrations (arsenic, fluoride, lithium, uranium) spanning two decades; resolved unit inconsistencies and geographic alignment issues.
- Developed complex SQL queries with multi-table joins for county-level aggregation and mineral co-occurrence analysis.
- Integrated database with Python (pandas, GeoPandas, Folium) to create choropleth maps identifying geographic patterns and contamination hotspots for public health research.

**Unsupervised Learning: Clustering & Association Rule Mining (Python)**

- Implemented and compared 6 clustering algorithms (K-means, Hierarchical Agglomerative Clustering, GMM, DBSCAN, Spectral, Graph-based) on 100K health/lifestyle records.
- Evaluated cluster quality using internal metrics (Silhouette, Davies-Bouldin, Calinski-Harabasz) and external metrics (ARI, NMI); visualized results with PCA projections.
- Conducted association rule mining on 541K retail transactions using Apriori and FP-Growth algorithms; discovered 79 actionable rules with lift up to 24 for product bundling strategies.
- Analyzed threshold sensitivity and identified high-confidence product trios for merchandising recommendations.

**Frito-Lay Employee Attrition Analysis (R)**

- Analyzed >1,000 HR records using R (tidyverse, caret) to identify drivers of turnover.
- Built Naive Bayes and KNN models achieving 78% accuracy; performed significance testing and cost modeling.

**Predicting Crab Age (R)**

- Modeled 25,000 crab measurements; engineered polynomial, interaction, and transformed features.
- Used stepwise and all-subsets regression to reduce multicollinearity, improving MAE to 0.15.
- Built R Shiny visual dashboards to display relationships between physiology and age.

**Housing Price Prediction (R)**

- Developed multivariate regression models with engineered features (log terms, interactions, normalization) achieving adjusted $R^2 = 0.87$.
- Cleaned and imputed 80+ variables; validated assumptions and generated Kaggle-ready predictions.

**Heart Disease Modeling (R & Python)**

- Analyzed >300,000 CDC BRFSS records to engineer health indices and identify predictors of heart disease.
- Compared models (KNN, LDA, QDA, logistic regression), achieving AUC > 0.80.
- Used ROC curves, confusion matrices, cross-validation, and feature importance for interpretability.

**Hospital Length of Stay Prediction (R)**

- Modeled patient LOS using linear regression, LASSO, kNN, and Random Forests.
- Identified operational and demographic predictors (infection risk, nursing ratios).

**Diabetes EDA & PCA/LDA (Python)**

- Visualized and analyzed Pima Indians dataset with pandas, scikit-learn, seaborn.
- Built PCA and LDA models and evaluated variance explained vs. class separation.

**Classification Model Pipeline Optimization (Python)**

- Built cross-validated pipelines for logistic regression, SVMs, tree-based models.
- Tuned hyperparameters using GridSearchCV and RandomizedSearchCV; assessed decision boundaries and calibration.